

Forecasting Daily WTI Crude Oil Price Direction Using Macroeconomic Features, Futures Positioning, Regime Modeling, and Machine Learning

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Abstract

This project evaluates whether daily WTI crude oil price direction can be forecast using macroeconomic indicators, CFTC/COT futures positioning data, HMM regime modeling, and machine learning classifiers. I compare Gradient Boosting, ELM, GRU, and FAVAR across pre-COT, COT-augmented, regime-probability, and regime-specific experiment versions. Classification performance remains modest, with most accuracies close to 50 percent and ROC-AUC values only slightly above random prediction. However, regime-specific modeling provides the clearest incremental value, especially when forecast probabilities are translated into a risk-controlled trading strategy. The strongest trading results come from the regime-specific GRU and FAVAR strategies, which outperform buy-and-hold during the final test period. Overall, the results suggest that alternative data and regime modeling may not produce highly accurate daily classifiers, but they can help generate more economically useful probability signals.

1 Introduction and Motivation

Forecasting crude oil prices is a difficult problem because oil markets are influenced by a mixture of macroeconomic conditions, geopolitical events, weather conditions, financial market behavior, supply and demand expectations, and short-term changes in investor positioning [1, 2]. Crude oil prices are also highly volatile because both supply and demand are relatively inelastic in the short run, meaning that shocks to expected supply or demand can produce large price changes [3]. Rather than forecasting the exact level of West Texas Intermediate (WTI) crude oil prices, this project focuses on predicting the daily direction of WTI price movements. The target is therefore framed as a binary classification problem of whether the next trading day's WTI price moves up or down. This framing is especially relevant for trading and risk management because many practical decisions depend less on the exact future price of an asset and more on whether the market is expected to move higher or lower.

The main objective of this project is to study whether machine learning models can extract useful directional signals from WTI crude oil data. I began with four forecasting models from an original classification task: Gradient Boosting, Extreme Learning Machine (ELM), Gated Recurrent Unit (GRU), and Factor-Augmented Vector Autoregression (FAVAR). These models represent different modeling families and trade-offs. Gradient Boosting is a nonlinear tree-based method designed for tabular prediction. ELM is a fast neural network style classifier with randomized hidden layer weights. GRU is a recurrent neural network designed to model sequential dependencies. FAVAR is a macroeconomic factor model that summarizes many correlated predictors into a smaller number of latent factors.

The original classification models provided a baseline for understanding how difficult the WTI direction prediction task is using price history and macroeconomic predictors alone. The macroeconomic variables came from the FRED-MD database. This is a large monthly macroeconomic dataset designed for empirical macroeconomic research using many U.S. economic time series [4]. Because the initial baseline results showed only modest directional forecasting performance, the results suggested that additional information sources or market structure modeling may be required to improve predictive performance.

To extend the original framework, I introduced two additional components. First, I added an alternative data source from the Commodity Futures Trading Commission (CFTC) Commitments of Traders (COT) reports for WTI futures. These reports provide weekly breakdowns of open interest positions in futures markets, including trader categories such as commercial and non-commercial participants [5]. The dataset captures trader positioning information, long and short positions, open interest, and changes in speculative positioning. The motivation for adding this dataset was to test whether futures market positioning contains incremental information about future oil price direction beyond standard macroeconomic indicators.

Second, I used a Hidden Markov Model (HMM) to identify latent market regimes. The motivation was that the relationship between predictors and oil price direction may change across different market environments, such as high-volatility drawdown periods versus lower-volatility trending periods. This idea is related to regime-switching models in time-series econometrics, where observed market behavior may be generated by multiple underlying latent states over time [6]. The regime framework was therefore used to test whether regime-aware forecasting models could better capture changing market dynamics.

The final stage of the project evaluated whether forecast probabilities could be translated into economic value through a probability-based trading strategy. Instead of only comparing classification accuracy, I tested whether model probabilities could guide position sizing and improve trading performance relative to a buy-and-hold benchmark. This distinction is important because a model with slightly better accuracy does not necessarily produce a better trading strategy. In financial markets, the timing of correct and incorrect predictions, the model’s confidence, the volatility environment, and drawdowns all affect realized trading performance.

Overall, the project is framed not as a search for a single best forecasting model, but as an exploration of trade-offs between model complexity, interpretability, alternative data, regime awareness, and economic usefulness. Different modeling approaches were evaluated not only based on statistical forecasting performance, but also based on their ability to generate economically meaningful trading signals.

2 Data and Preprocessing

2.1 Forecasting Target Construction and Price Features

The core forecasting task uses WTI crude oil price data to predict the next-day direction of price movement [7]. Rather than forecasting the exact future price level, the project focuses on directional classification because the primary objective is to support trading and risk-management decisions. The target variable was constructed as a binary label based on the following day’s return. If the next trading day’s WTI price increased, the target was assigned a value of 1. If the next trading day’s WTI price decreased or did not increase, the target was assigned a value of 0. This means that all model features at time t were used to predict the direction of the price movement from time t to time $t + 1$.

The binary target variable was defined as:

$$y_t = \begin{cases} 1 & \text{if } r_{t+1} > 0 \\ 0 & \text{otherwise} \end{cases}$$

where r_{t+1} represents the next trading day return of WTI crude oil prices.

This target construction is important because it avoids using future information in the model inputs. In time-series forecasting, even small forms of look-ahead bias can make results appear substantially better than they would be in a realistic trading environment. For that reason, the data was kept in chronological order throughout the project, and all model training, validation, and testing procedures preserved the temporal ordering of observations.

The baseline models also derived features from the WTI price data. The original implementation used 30 lagged price observations which means that each prediction was based on a 30-day lookback window of

recent market behavior. This lookback structure was especially important for the GRU model which requires sequential inputs. However, the same lagged framework was used across all the models as well.

2.2 Macroeconomic Predictors

The original forecasting pipeline used macroeconomic variables from the FRED-MD dataset [4]. FRED-MD is a large monthly macroeconomic database designed for empirical macroeconomic forecasting and includes variables from multiple categories of U.S. economic activity. The macroeconomic variables used in this project included output and income measures, labor-market variables, interest rates, credit spreads, exchange rates, commodity-price indicators, and sentiment measures. The motivation for including these predictors is that crude oil prices are closely linked to broader economic conditions. Stronger economic activity can increase demand expectations for energy, while interest rates, credit conditions, inflation, and exchange rates can influence both commodity demand and financial flows into oil-related assets.

The specific FRED-MD variables used in the forecasting pipeline are listed in Table 1. Together, these variables provide information about economic activity, financial conditions, inflationary pressure, exchange-rate movements, and expectations.

Table 1: FRED-MD macroeconomic variables used in the forecasting pipeline

Variable	Description	Potential relevance to oil-price direction
RPI	Real Personal Income	Measures household income and broad economic strength, which can influence energy demand.
W875RX1	Real Personal Income Less Transfer Receipts	Captures underlying income excluding government transfers, providing a cleaner measure of private-sector income.
CMRMTSPLx	Real Manufacturing and Trade Industries Sales	Measures real sales activity in manufacturing and trade sectors, which may be linked to industrial and transportation demand for energy.
IPFPNSS	Industrial Production: Final Products and Nonindustrial Supplies	Measures industrial output, a direct proxy for energy-intensive economic activity.
USWTRADE	All Employees: Wholesale Trade	Labor-market indicator for wholesale trade activity, which may reflect goods movement and business-cycle conditions.
USTRADE	All Employees: Retail Trade	Labor-market indicator for retail trade activity and consumer-facing economic conditions.
BUSLOANS	Commercial and Industrial Loans	Measures business credit conditions, which can affect investment, production, and energy demand expectations.
CONSPI	Nonrevolving Consumer Credit to Personal Income	Captures consumer credit conditions relative to income and may reflect household financial stress or spending capacity.
S&P 500	S&P 500 Stock Price Index	Broad financial-market indicator that captures risk appetite and expectations about economic growth.

Variable	Description	Potential relevance to oil-price direction
S&P PE ratio	S&P 500 Price-Earnings Ratio	Measures equity-market valuation and investor expectations about future earnings and growth.
FEDFUNDS	Effective Federal Funds Rate	Short-term policy rate that affects financing conditions, the dollar, and macroeconomic demand.
TB3MS	3-Month Treasury Bill Rate	Short-term interest rate reflecting monetary conditions and liquidity.
TB6MS	6-Month Treasury Bill Rate	Short-term interest rate reflecting expectations about near-term monetary policy.
GS1	1-Year Treasury Rate	Interest-rate measure capturing short-to-intermediate monetary expectations.
GS5	5-Year Treasury Rate	Intermediate-term Treasury rate linked to growth and inflation expectations.
GS10	10-Year Treasury Rate	Long-term Treasury rate capturing macroeconomic expectations and discount-rate conditions.
AAA	Moody's Seasoned Aaa Corporate Bond Yield	High-grade corporate borrowing rate, reflecting credit conditions for lower-risk firms.
BAA	Moody's Seasoned Baa Corporate Bond Yield	Lower-rated corporate borrowing rate, reflecting riskier credit conditions and financial stress.
TB3SMFFM	3-Month Treasury Bill Minus Federal Funds Rate	Short-term interest-rate spread that captures money-market and policy-rate conditions.
TB6SMFFM	6-Month Treasury Bill Minus Federal Funds Rate	Short-term spread reflecting expectations about near-term rate movements.
T1YFFM	1-Year Treasury Minus Federal Funds Rate	Yield-spread measure capturing expectations about future monetary policy.
T5YFFM	5-Year Treasury Minus Federal Funds Rate	Intermediate yield-spread measure related to growth and inflation expectations.
T10YFFM	10-Year Treasury Minus Federal Funds Rate	Yield-curve spread often associated with recession risk and long-term macroeconomic expectations.
AAAFFM	Aaa Corporate Bond Yield Minus Federal Funds Rate	Credit spread measure for high-grade firms, reflecting financial conditions relative to policy rates.
BAAFFM	Baa Corporate Bond Yield Minus Federal Funds Rate	Credit spread measure for riskier corporate debt, often associated with financial stress.
EXSZUSx	Switzerland/U.S. Foreign Exchange Rate	Exchange-rate variable capturing international currency movements and safe-haven dynamics.
EXJPUSx	Japan/U.S. Foreign Exchange Rate	Exchange-rate variable reflecting global currency and financial conditions.

Variable	Description	Potential relevance to oil-price direction
EXUSUKx	U.S./U.K. Foreign Exchange Rate	Exchange-rate variable capturing international financial and trade conditions.
EXCAUSx	Canada/U.S. Foreign Exchange Rate	Exchange-rate variable relevant because Canada is a major oil producer and exporter.
PPICMM	Producer Price Index: Metals and Metal Products	Commodity-price indicator that captures broader input-cost and inflation pressures.
UMCSENTx	University of Michigan Consumer Sentiment Index	Measures consumer expectations and confidence, which can relate to demand expectations and risk sentiment.

2.3 FRED-MD Transformations and Macro Alignment

The FRED-MD dataset includes transformation codes that indicate how each macroeconomic series should be transformed before modeling [4]. These transformations are designed to improve stationarity and comparability across variables. Stationarity is important because many time-series forecasting methods perform poorly when variables contain strong trends, changing variance, or persistent nonstationary behavior.

The FRED-MD transformation codes include several common transformations. A transformation code of 1 indicates that the variable is used in levels with no transformation. Codes 2 and 3 represent first and second differences, which are used to remove trends and stronger forms of nonstationarity. Code 4 applies a logarithmic transformation to stabilize variance. Codes 5 and 6 apply first and second differences to logarithmic values, which are commonly used to model growth rates and changes in growth rates. Transformation code 7 represents relative changes or percent-change style transformations.

For example, income or industrial production variables are often transformed using log differences so that the forecasting models use growth rates rather than raw levels. These transformations help place variables measured on very different scales into forms that are more comparable and statistically stable for forecasting.

A key preprocessing challenge is that FRED-MD variables are monthly, while WTI prices are observed daily. In addition, macroeconomic data is not known immediately at the beginning of the labeled month. To reduce the risk of look-ahead bias, the macroeconomic data was shifted backward by one month before being aligned with the daily WTI data. For example, if the model was making a prediction in November, it would have access to the macroeconomic data from October. This reflects the idea that a forecaster would only be able to use macroeconomic information after it would realistically have been released.

2.4 Data Splits, Cross-Validation, and Scaling

The data was split chronologically rather than randomly. Random shuffling was intentionally avoided because shuffling would allow future observations to appear in the training set while earlier observations appear in validation or test sets, creating unrealistic information leakage. The original VIP pipeline used historical FRED-MD vintages from 1999-08 to 2014-12 for training and historical vintages from 2015-01 to 2024-12 for validation. In the final-project extension, the available modeling sample changed after adding the CFTC/COT data source, which began later than the original macroeconomic dataset. The final out-of-sample test period was 2025-01-01 and ended on 2025-10-31.

During model selection, the training data was split using 5-fold time-series cross-validation. Unlike standard cross-validation, time-series cross-validation preserves temporal ordering by ensuring that validation observations occur after training observations. This is more appropriate for forecasting because future data should never be used to predict the past.

Standard scaling was used for normalization. The scaler was fit only on the training data and then applied to validation and test sets. This avoids data leakage because fitting the scaler on the full dataset

would allow information from future validation or test periods to influence the transformation of earlier training observations. The final test set was not used during model development, hyperparameter tuning, or trading-strategy optimization and was reserved exclusively for final out-of-sample evaluation.

3 Baseline Forecasting Models

Before adding CFTC/COT data, HMM regimes, or trading strategy evaluation, I first implemented the four required classification models on the original WTI direction forecasting task. These original models served as an informative baseline for the project and helped establish how difficult the prediction problem was using only the initial WTI price history and macroeconomic feature set. The four models were chosen because they represent different modeling families: a nonlinear tree ensemble (Gradient Boosting), a randomized neural-network-style classifier (ELM), a recurrent deep learning model (GRU), and a macroeconomic factor model (FAVAR). Comparing them made it possible to evaluate not only predictive performance, but also the trade-offs between flexibility, interpretability, temporal structure, and robustness.

The baseline models were evaluated using standard classification metrics: accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Accuracy measures the overall fraction of correctly classified days. Precision measures how often the model was correct when it predicted an up day. Recall measures how many actual up days the model successfully identified. F1-score combines precision and recall into one metric. ROC-AUC measures the model’s ability to distinguish between up and down days across probability thresholds. Because this is a financial time-series classification problem, the confusion matrix is especially important. A model can appear reasonable by accuracy or F1-score while still failing to identify one of the two market directions.

For all reported metrics in this section, the positive class is an “up” day, meaning that WTI increased on the following trading day.

3.1 Gradient Boosting

Gradient Boosting is a nonlinear tree-based ensemble method. It builds a sequence of decision trees, where each new tree attempts to correct the mistakes of the previous trees [8]. This model belongs to the supervised machine learning family of boosting algorithms and is well suited for tabular prediction problems because it can capture nonlinear relationships and interactions among predictors without requiring a manually specified functional form.

The architecture of Gradient Boosting is an additive collection of small decision trees. Each tree makes a simple set of rule-based splits, such as whether volatility, momentum, or a macroeconomic variable is above or below a certain threshold. The model then combines the trees into one final prediction. Instead of fitting all trees independently, Gradient Boosting fits them sequentially. Each new tree focuses on the errors left by the previous trees, so the ensemble gradually improves by paying more attention to observations that were harder to classify [8]. In this project, the final output of the boosted trees was converted into a predicted probability that the next trading day would be an up day.

The main assumption behind Gradient Boosting is not a strict statistical distributional assumption, but rather the idea that combining many weak learners can approximate a useful prediction function. In this project, the implicit assumption was that lagged price features and macroeconomic predictors contain nonlinear patterns that can help distinguish future up days from down days. However, because crude oil returns are noisy and non-stationary, a flexible boosted model can also overfit relationships that do not persist out of sample.

In the original Gradient Boosting notebook, the model was tuned using cross-validation. The best configuration used a learning rate of 0.05, a maximum tree depth of 2, 200 boosting iterations, a minimum leaf size of 60, and L2 regularization of 0.25. The shallow tree depth helped control overfitting, while the minimum leaf size and L2 regularization further limited the model’s flexibility. This was important because the goal was not to fit historical noise, but to find a signal that could generalize to the held-out test period.

On the original test set, Gradient Boosting achieved an accuracy of approximately 0.542 and a ROC-AUC of approximately 0.545. Its precision was approximately 0.524, recall was approximately 0.822, and F1-score was approximately 0.640.

Table 2: Gradient Boosting baseline confusion matrix.

Actual / Predicted	Predicted Down	Predicted Up
Actual Down	29	80
Actual Up	19	88

The model correctly identified many actual up days, but it also misclassified many down days as up days. This means the model had a strong upward prediction bias. The high recall for up days therefore needs to be interpreted carefully. It did not necessarily mean the model had a strong understanding of oil price direction; rather, it often predicted the up class. Gradient Boosting was one of the stronger original baselines by ROC-AUC, but the confusion matrix showed that its directional predictions were not balanced.

3.2 Extreme Learning Machine

The Extreme Learning Machine is a neural-network-style classifier with a single hidden layer. Unlike a standard neural network, the hidden-layer weights are randomly initialized and then held fixed. This means that no backpropagation is used. The output layer is estimated directly, usually with a regularized linear solution [9]. This makes ELM much faster to train than a fully optimized neural network while still allowing nonlinear transformations of the input features.

The ELM architecture has three main parts: an input layer, a hidden layer, and an output layer. The input layer receives the scaled macroeconomic and price-based predictors. These inputs are then passed through a hidden layer with randomly generated weights and a nonlinear activation function. This hidden layer creates a large transformed representation of the original feature set. Unlike a standard neural network, the hidden-layer weights are not updated through backpropagation [9]. Only the final output-layer weights are learned. The output layer then maps the transformed hidden representation into a classification prediction for the next-day WTI direction.

ELM belongs to the family of randomized neural models. Its main assumption is that a sufficiently large random hidden layer can transform the input data into a feature space where the classes are easier to separate. The model is flexible and computationally efficient, but its performance can be sensitive to the number of hidden neurons, the activation function, the random weight initialization, and the amount of regularization. Because the hidden layer is random, the model may also be less stable than a fully trained neural network.

In the original ELM notebook, the best model used a tanh activation function, Xavier-style weight initialization, 1,650 hidden neurons, and a regularization parameter of 1e-06. The hidden-layer weights and biases were randomly initialized and then held fixed, so the model did not update these weights through backpropagation. Instead, after the input features were passed through the hidden layer, the model produced a hidden activation matrix H . I then estimated the output-layer weights directly using a ridge regression solution:

$$\beta = (H^T H + \alpha I)^{-1} H^T y.$$

Here, H is the hidden-layer representation, y is the binary WTI direction target, α is the regularization parameter, and β represents the learned final-layer weights. This means that the only learned part of the ELM was the final output layer. After calculating $H\beta$, I passed the raw score through a sigmoid function to obtain the predicted probability of an up day.

On the original test set, ELM achieved an accuracy of approximately 0.528 and a ROC-AUC of approximately 0.532. Its precision was approximately 0.518, recall was approximately 0.538, and F1-score was

approximately 0.528.

Table 3: ELM baseline confusion matrix.

Actual / Predicted	Predicted Down	Predicted Up
Actual Down	57	53
Actual Up	49	57

The ELM confusion matrix shows that the model did not collapse into predicting only one class. It correctly identified 57 down days and 57 up days, while misclassifying 53 down days and 49 up days. This indicates that the model produced relatively balanced predictions across the two classes. However, the overall predictive signal was still weak, with accuracy only slightly above 50% and ROC-AUC only slightly above random classification. This suggests that the random nonlinear hidden-layer representation captured some structure in the original feature set, but not enough to produce a strong standalone directional forecasting model.

3.3 GRU

The Gated Recurrent Unit is a recurrent neural network architecture designed for sequential data. GRUs use gating mechanisms to decide how much past information to retain and how much new information to update at each time step. This makes them useful for problems where recent observations may influence future outcomes. In this project, the GRU was included because WTI prices and macro-financial variables may contain temporal dependencies that are not fully captured by static tabular models.

The GRU architecture processes the input data as an ordered sequence rather than as independent rows. At each time step, the model receives the current feature vector and updates a hidden state, which acts as a memory of recent information [10]. The update gate controls how much information from the previous hidden state should be carried forward, while the reset gate controls how much of the past should be ignored when incorporating the current observation. After the GRU processes the full lookback window, the final hidden state is passed into a classification layer, which outputs the predicted probability of an up day.

The GRU belongs to the deep learning family of sequence models. Its main assumption is that the order of historical observations matters and that a learned hidden state can summarize useful information from recent time steps. This is a reasonable motivation for financial time series, but it also creates challenges. Daily oil direction labels are noisy, the signal-to-noise ratio is low, and the available training sample is limited compared with many deep learning applications. As a result, a GRU can overfit, fail to generalize, or collapse toward predicting one class if the directional signal is weak.

In the original GRU notebook, the model used a 30-day lookback window of WTI prices as the sequential input. Each training example contained a price sequence with shape 30×1 , representing the previous 30 daily WTI prices. The macroeconomic variables were not passed through the recurrent layer as a sequence. Instead, the GRU first processed the 30-day price sequence and produced a final hidden state summarizing recent price behavior. This hidden state was then concatenated with the current macroeconomic feature vector, which contained 31 macroeconomic predictors. The combined representation was passed through a final linear layer to produce a single logit for binary classification.

I used this design because the inputs had different temporal structures. The WTI price history was naturally sequential, so the GRU was used to learn a compact representation of the recent 30-day price path. The macroeconomic variables were lower-frequency predictors that described the current economic environment after the reporting lag adjustment. Rather than forcing these macro variables through the recurrent layer as if they contained daily sequence dynamics, I concatenated them with the final GRU hidden state before the output layer. This allowed the model to combine short-term oil price dynamics with broader macroeconomic

conditions in the final classification step.

In the final selected GRU implementation, the model used one recurrent layer with a hidden size of 64 and a batch size of 64. The hidden size controls the dimension of the GRU’s internal memory representation. A hidden size of 64 provided enough capacity to summarize the 30-day price sequence without making the model overly large for the available data. Using one recurrent layer kept the model relatively simple and helped reduce overfitting, which was important because daily oil direction is a noisy target and the data sample was small. The batch size of 64 controlled how many training examples were used in each optimization step. The model was trained with the Adam optimizer using a learning rate of 0.0005, which controlled the size of each parameter update during training. The loss function was binary cross-entropy with logits, which is appropriate for a two-class direction prediction problem because the model outputs a single raw score, or logit, for the probability of an up day. The predicted logits were converted into probabilities, and a classification threshold of 0.50 was used to assign each observation to either the up or down class.

On the original test set, the GRU achieved an accuracy of approximately 0.491 and a ROC-AUC of approximately 0.501. Its precision was approximately 0.491, recall was 1.000, and F1-score was approximately 0.659.

Table 4: GRU baseline confusion matrix.

Actual / Predicted	Predicted Down	Predicted Up
Actual Down	0	110
Actual Up	0	106

Although the recall and F1-score may appear acceptable, the confusion matrix shows that the model predicted every test observation as an up day. This is a class collapse problem. The model did not learn to distinguish between up and down days; it simply assigned all observations to one class. This result is important because it shows why confusion matrices are necessary in addition to summary metrics. The GRU’s performance below 50% accuracy and its class collapse suggest that a more complex deep learning architecture does not automatically improve forecasting performance when the financial signal is weak and unstable. This does not necessarily mean that GRUs are inappropriate for oil price forecasting in general. Rather, the result may reflect the limited sample size, the low signal-to-noise ratio of daily direction labels, and the difficulty of training a deep sequence model on relatively short and noisy financial time series.

3.4 FAVAR

The Factor-Augmented Vector Autoregression model takes a different approach from the other classifiers. Instead of directly fitting a flexible classifier to all predictors, FAVAR summarizes a large number of macroeconomic variables into a smaller set of latent factors. These factors capture common movements across the macroeconomic dataset. The model then uses autoregressive structure to model how these factors evolve over time and how they relate to the target variable [11].

The FAVAR pipeline has two main stages. First, the macroeconomic predictor set is reduced into a smaller number of common factors, usually through a dimensionality-reduction method such as principal component analysis. These factors are intended to summarize broad economic movements shared across many correlated variables, such as production, income, interest rates, credit conditions, and inflation. Second, a vector autoregression is fit using the extracted factors and lagged information. This allows the model to use time-series relationships among the factors to generate a forecast. The forecast is then converted into a directional prediction for WTI.

FAVAR belongs to the family of macroeconomic time-series models. Its main assumption is that a smaller number of unobserved factors can summarize the information contained in many correlated macroeconomic

variables. This is useful when the predictor set is large and many variables move together because of shared economic conditions [11]. FAVAR is more interpretable than a black-box model like a neural network because the factors can be understood as broad macroeconomic forces. However, it may struggle if the relationship between macro factors and oil direction is nonlinear, unstable, or regime-dependent.

In the original FAVAR notebook, the best validation setup used five factors and four VAR lags. The factor structure reduced the dimensionality of the macroeconomic feature set, while the lag structure allowed the model to incorporate temporal dynamics. This made FAVAR a natural benchmark for the original macro-driven forecasting task because it directly reflects the idea that oil prices may respond to broad macroeconomic conditions rather than isolated individual indicators.

On the original test set, FAVAR achieved an accuracy of approximately 0.551, making it the strongest original baseline by test accuracy. Its precision was approximately 0.54, recall was approximately 0.65, and F1-score was approximately 0.59.

Table 5: FAVAR baseline confusion matrix.

Actual / Predicted	Predicted Down	Predicted Up
Actual Down	49	60
Actual Up	37	70

The FAVAR model produced a more balanced confusion matrix than Gradient Boosting and avoided the class collapse problem observed in the GRU. It still misclassified many down days as up days, but it identified more down days than the Gradient Boosting model. This made FAVAR one of the most useful original baselines because it combined relatively strong test accuracy with a more interpretable macroeconomic structure.

3.5 Baseline Model Comparison

The original baseline results showed that predicting daily WTI direction is a difficult task. All four models produced test accuracies near 50 to 55 percent, and the confusion matrices showed important weaknesses that were not always visible from summary metrics alone. FAVAR and Gradient Boosting were the strongest original models by test accuracy, while ELM provided a relatively balanced but weak nonlinear benchmark. The GRU struggled the most because it collapsed into predicting a single class on the test set.

Table 6: Original baseline model comparison.

Model	Family	Accuracy	Precision	Recall	F1	Main Observation
Gradient Boosting	Tree ensemble	0.542	0.524	0.822	0.640	Up-class bias
ELM	Randomized neural model	0.528	0.518	0.538	0.528	Balanced but weak
GRU	Recurrent neural network	0.491	0.491	1.000	0.659	Class collapse
FAVAR	Macro factor model	0.551	0.538	0.654	0.591	Best baseline accuracy

The four models illustrate different trade-offs. Gradient Boosting was flexible and handled nonlinear feature interactions, but it leaned heavily toward the up class. ELM was fast and more balanced, but its overall predictive power was limited. GRU was theoretically attractive because it could model sequences, but in practice it failed to generalize and collapsed into one class. FAVAR was less flexible than the machine learning models but provided the best original accuracy and a clearer macroeconomic interpretation.

Overall, the original baseline results suggested that the macroeconomic and price-based feature set contained some directional information, but the signal was weak. This motivated the next stage of the project: adding alternative trader-positioning data from the CFTC/COT reports, using HMM regimes to represent changing market conditions, and evaluating whether the resulting forecast probabilities could produce economic value through a trading strategy.

4 Alternative Data: CFTC/COT

After establishing the original baseline forecasting models, I extended the dataset by adding futures market positioning data from the Commodity Futures Trading Commission (CFTC) Commitments of Traders (COT) reports [5]. The motivation for adding this alternative dataset was that macroeconomic variables and lagged prices may not fully capture how market participants are positioned in WTI futures markets. Since crude oil is both a physical commodity and a heavily traded financial asset, trader positioning may contain information about expectations, hedging demand, speculative pressure, and changes in market sentiment. If speculative positioning changes before prices fully adjust, these variables may provide an early signal about shifts in market expectations.

4.1 Dataset Source and Frequency

The COT reports provide weekly information about futures market open interest and the positions held by different trader categories. For WTI futures, the data includes long and short positions for commercial and non-commercial traders, along with total open interest. Commercial traders are typically associated with producers, consumers, or hedgers that use futures markets to manage exposure to the physical commodity. Non-commercial traders are generally more associated with speculative positioning. Because the data is reported weekly, it has a lower frequency than the daily WTI price data used for the forecasting target.

The key distinction between the COT dataset and the macroeconomic predictors is that COT data is market-positioning based rather than macroeconomic. While FRED-MD variables describe broad economic conditions, COT variables describe how futures market participants are positioned. This makes the dataset a potentially useful complement to the original macroeconomic feature set.

4.2 Feature Construction

The raw COT variables were cleaned and transformed into features designed to summarize trader positioning. In addition to raw long and short positions, I created features such as net positioning, long-short ratios, open interest measures, and changes in positioning over time. Net positioning was especially important because it summarizes whether a trader group is more heavily long or short. Long-short ratios provide a relative measure of directional imbalance, while open interest captures the overall level of market participation.

Table 7: CFTC/COT positioning features added to the forecasting pipeline.

Feature Type	Interpretation
Open interest	Measures total outstanding futures contracts and captures overall market participation.
Commercial long/short positions	Capture hedging activity from commercial market participants.
Non-commercial long/short positions	Capture speculative positioning from non-commercial traders.
Net positioning	Measures long positions minus short positions for a trader group.
Long-short ratio	Measures relative directional imbalance between long and short exposure.
Changes in positioning	Capture week-to-week shifts in trader behavior and sentiment.

These engineered features were intended to capture changes in speculative pressure and market positioning that may not be visible from price history or macroeconomic variables alone. For example, a large increase in non-commercial net long positioning may indicate stronger speculative demand for crude oil exposure, while a sharp decrease may indicate reduced conviction or a shift toward bearish positioning. However, these interpretations should be viewed as potential signals rather than deterministic relationships.

4.3 Exploratory Analysis

Before adding the COT features to the forecasting models, I examined the structure of the positioning data. This step was important because the COT variables are measured on very different scales from the macroeconomic and price-based variables. Raw position counts can also change over time as futures market participation changes. For this reason, features such as ratios, net positions, and changes in positions were useful because they provide more interpretable measures of trader imbalance than raw contract counts alone.

The exploratory analysis focused on how positioning evolved over time, how open interest changed, and whether the engineered positioning features were highly concentrated or skewed.

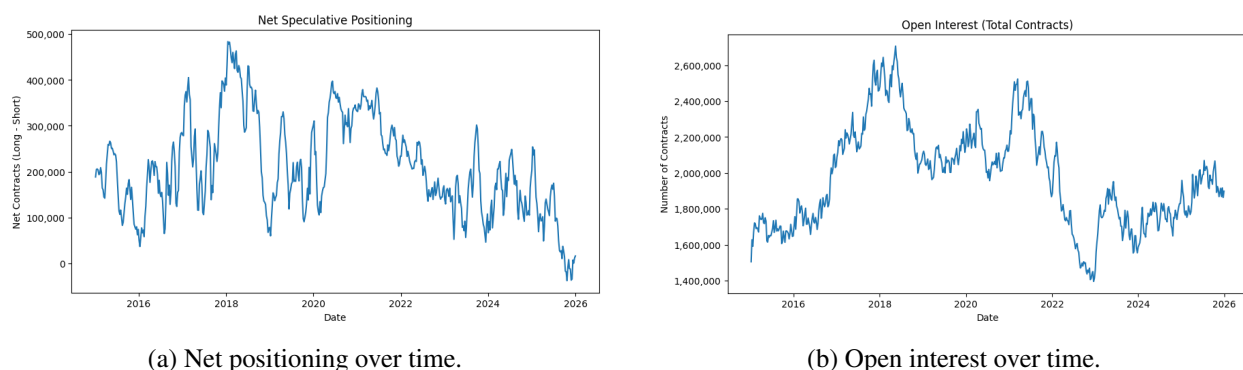


Figure 1: Exploratory analysis of CFTC/COT WTI positioning data. Net positioning captures directional imbalance, while open interest captures the overall level of futures market participation.

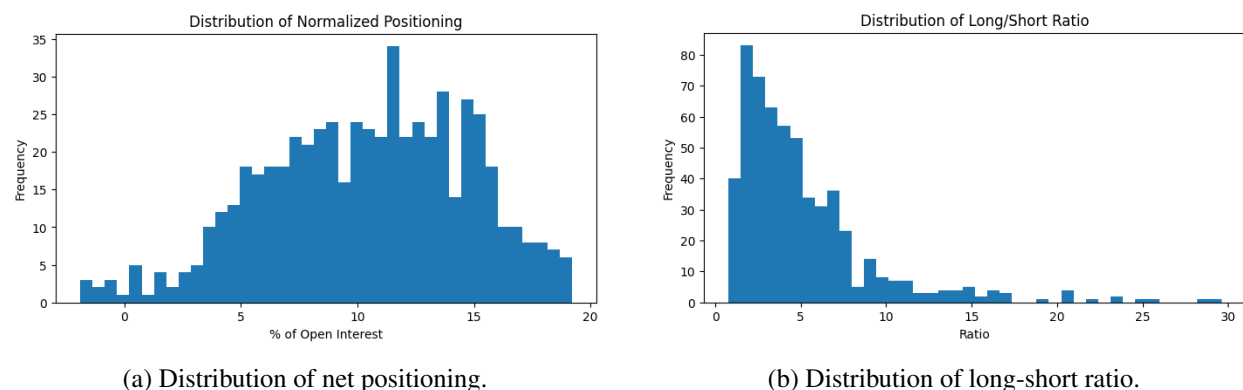


Figure 2: Distributional checks for engineered COT features. These plots helped evaluate whether positioning variables were skewed or concentrated before scaling and model training.

4.4 Integration into the Forecasting Pipeline

Because the COT reports are weekly while WTI prices are observed daily, the COT data had to be converted from a weekly series into a daily feature set before it could be merged into the forecasting pipeline. I handled

this by resampling the COT data to a daily frequency and forward-filling each weekly observation until the next report became available. This means that the COT variables were constant within each reporting week because no new positioning information was released between reports.

An additional preprocessing challenge is that the COT reports are constructed using trader positions as of Tuesday but are not publicly released until Friday. Without adjusting for this reporting delay, the forecasting models could incorrectly use information before it would realistically have been available, creating look-ahead bias.

To account for this timing structure, the daily COT features were shifted forward by three days after resampling. This adjustment ensures that Tuesday positioning data only becomes available to the forecasting models starting on Friday, which more accurately reflects the true timing of information availability in financial markets.

This integration changed the usable sample period. The original baseline pipeline used historical FRED-MD vintages beginning in 1999, but the merged COT-enhanced dataset could only begin once both the WTI/macroeconomic variables and COT positioning variables were available. As a result, the final COT-augmented modeling dataset began on 2015-05-21 and extended through the final test period ending on 2025-10-31. The final number of rows after combining the data sources is 2749. This created an important trade-off: adding futures-positioning data introduced a potentially useful alternative signal, but it reduced the amount of historical data available for model training and validation.

4.5 Limitations of the COT Data

The COT dataset has several limitations. First, it is reported weekly, while the forecasting target is daily. This means the positioning data may not capture short-term changes that occur between report dates. Second, the reports are released with a delay, so timing and availability must be handled carefully to avoid look-ahead bias. Third, trader positioning is noisy and does not necessarily imply future price direction. A large speculative long position may indicate bullish sentiment, but it may also indicate crowded positioning that could reverse if market conditions change.

Despite these limitations, the COT dataset was valuable because it introduced a different type of information from the original macroeconomic predictors. The purpose of adding it was not to assume that positioning would always improve forecasts, but to test whether futures market positioning provided incremental signal when combined with the original WTI and macroeconomic features.

5 Regime Modeling with Hidden Markov Models

After adding the CFTC/COT positioning variables, I introduced a Hidden Markov Model (HMM) to identify market regimes in the WTI dataset. The motivation for this step was that the relationship between predictors and oil price direction may not be stable across all market environments. For example, momentum, volatility, macroeconomic variables, and trader positioning may have different meanings during a high-volatility drawdown period than during a more stable trending market. A single classifier trained across the entire sample may therefore struggle if it assumes that the same predictor relationships hold across all periods.

A Hidden Markov Model is a probabilistic model that assumes the observed data are generated by a set of unobserved, or hidden, states [12]. In this project, the hidden states were interpreted as latent WTI market regimes. The model does not directly observe whether the market is in a high-volatility, low-volatility, or positioning-driven state. Instead, it estimates these regimes from patterns in the observed features. Each day is assigned to the regime that is most likely given the observed feature values and the transition structure learned by the model. This idea is related to regime-switching time-series models, where observed financial or macroeconomic data may behave differently across latent states [6].

The purpose of the HMM was not to directly predict the next-day WTI direction. Instead, it was used

as a feature engineering and model-structuring tool. Once the regimes were estimated, I used them in two ways. First, I included the estimated regime probabilities as additional features in the classification models. This allowed the classifiers to condition their predictions on the current estimated market environment. Second, I trained regime-specific versions of the models, where the classification problem was separated by regime. This tested whether different market states had different predictor relationships that could be better captured by separate models.

Besides the new CFTC/COT data, I also derived price-based features. Specifically, I created rolling 20 and 60 day volatility features, a drawdown measure, 20 day price momentum based on price percentage changes, and a trend column that compares the 20 and 100 day rolling mean price.

The first HMM implementation used the full feature set after dropping the target variable, macro month, and price level. The features were standardized and then reduced using principal component analysis before fitting the HMM. This version initially appeared reasonable in-sample, but it produced poor regime diversity out of sample. Most validation and test observations were assigned to the same regime. A single regime would be assigned to the entire year. This made the regime labels less useful because they did not provide much additional information about changing market conditions.

To address this issue, I redesigned the HMM feature construction by separating the predictors into three groups: macroeconomic features, market-based WTI features, and CFTC/COT positioning features. This mattered because the three feature groups operated at different timescales. The WTI market features changed daily, the CFTC/COT variables were reported weekly and then aligned to the daily dataset, and the macroeconomic variables were monthly and shifted to account for reporting delays. If these groups were combined directly after scaling, the HMM could be dominated by the feature block whose variation was most frequent or most emphasized after PCA, rather than learning regimes that balanced macroeconomic conditions, trader positioning, and daily oil-market behavior.

To reduce this problem, I applied PCA separately within each feature group and then down-weighted the resulting PCA features according to the data interval. This helped prevent the regime model from being driven too heavily by one timescale of information. The goal was to create regimes that reflected a balanced combination of slow-moving macroeconomic conditions, medium-frequency CFTC/COT positioning, and high-frequency WTI market behavior.

The final HMM used three regimes, three PCA components, a diagonal covariance structure, a minimum covariance value of 0.01, a transition-matrix prior of 10.0, and MAP estimation. I selected this specification because it produced more interpretable and more balanced regime assignments across the training and validation periods. The diagonal covariance structure kept the regime model relatively simple by assuming that the PCA components had regime-specific variances but not unrestricted covariance relationships. This was useful because the HMM was not intended to be a standalone forecasting model. Instead, its value came from converting the market, macroeconomic, and CFTC/COT variables into a compact regime representation that summarized the current market environment.

5.1 Regime Interpretation

The regime interpretation step was important because HMM labels are not meaningful by themselves. The model only returns numerical state labels such as regime 0, regime 1, and regime 2. To give these labels economic meaning, I compared average market behavior and CFTC/COT positioning characteristics across regimes. Table 8 reports the average values of selected WTI market and COT variables within each regime.

Table 8: Average market and COT characteristics by HMM regime.

Regime	Returns	Vol 20	Vol 60	Momentum 20	Drawdown	Net Position %	Flow	Long/Short Ratio	Open Interest
0	0.000244	0.036401	0.037902	-0.022939	-0.188552	8.221	-2853.447	3.025	1,940,832
1	0.000545	0.015873	0.017463	0.028293	-0.041111	13.131	3199.445	8.571	2,221,479
2	0.001806	0.020477	0.021901	0.034581	-0.049067	14.275	-1908.411	5.890	2,201,333

Based on these characteristics, regime 0 appears to represent a more stressed or weaker oil market environment. It had the highest 20-day and 60-day volatility, the most negative 20-day momentum, and the largest average drawdown. It also had the lowest net positioning percentage and the lowest long-short ratio. These characteristics suggest that regime 0 captured periods when recent price behavior was weaker, volatility was elevated, and trader positioning was less bullish.

Regime 1 appears to represent a more stable and positive momentum environment. It had the lowest volatility, positive 20-day momentum, a much smaller average drawdown, and the highest long-short ratio. Its positive flow value also suggests that trader positioning was improving on average during this regime. Compared with regime 0, regime 1 therefore looked like a healthier or more trend-supportive market state.

Regime 2 was the strongest regime by average daily returns and also had positive momentum and high net positioning. However, it differed from regime 1 because it had a negative average flow and a lower long-short ratio than regime 1. This suggests that regime 2 may have captured a historically unusual market state where returns and positioning levels were strong, but changes in positioning were not as favorable. As shown in the yearly distribution table, this regime was heavily concentrated in specific years, especially around the 2020–2022 period.

To summarize these interpretations, Table 9 gives a qualitative description of each regime.

Table 9: Economic interpretation of HMM regimes.

Regime	Interpretation	Main Characteristics
0	Stressed / volatile oil market environment	Highest volatility, weakest momentum, deepest drawdowns, lower net positioning percentage, and lowest long-short ratio. This regime appears to capture weaker and more unstable WTI market conditions.
1	Stable / positive momentum environment	Lowest volatility, positive momentum, smaller drawdowns, positive positioning flow, and highest long-short ratio. This regime appears to capture a more favorable and trend-supportive market environment.
2	Event-driven / historically concentrated positive-return regime	Highest average returns and positive momentum, but negative positioning flow. This regime was concentrated around specific historical periods and disappeared in the final test period.

5.2 Yearly Regime Distribution

I also examined the normalized distribution of regimes by year. This was useful because it showed whether the HMM states were spread evenly over time or concentrated in particular market periods. Table 10 reports the share of each year assigned to each regime.

The yearly distribution supports the interpretation that the HMM was identifying time-varying market environments rather than assigning labels randomly. Regime 0 dominated 2015, 2016, 2019, 2022, 2024,

and 2025. Regime 1 dominated 2017 and 2018 and remained present in several later years. Regime 2 was concentrated in 2020, 2021, and part of 2022. This timing suggests that regime 2 may have captured an unusual oil-market period around the COVID-19 shock and subsequent recovery, rather than a regime that appeared consistently throughout the sample.

Table 10: Normalized yearly distribution of HMM regimes.

Year	Regime 0	Regime 1	Regime 2
2015	0.796	0.204	0.000
2016	0.697	0.303	0.000
2017	0.099	0.901	0.000
2018	0.163	0.837	0.000
2019	0.595	0.405	0.000
2020	0.466	0.064	0.470
2021	0.000	0.000	1.000
2022	0.683	0.158	0.158
2023	0.545	0.455	0.000
2024	0.672	0.328	0.000
2025	0.815	0.185	0.000

The disappearance of regime 2 in 2023, 2024, and 2025 is an important limitation. It shows that regime models can identify historically meaningful states, but those states may not always reappear in future periods. This matters for regime-specific modeling because a classifier trained for a rare or historically concentrated regime may have limited usefulness if that regime does not occur in the test period. In my final test set, only regimes 0 and 1 appeared, meaning that the regime-specific models had to rely mostly on those two states during out-of-sample evaluation.

Overall, the HMM extension was useful because it introduced market-state information into the forecasting pipeline. The initial HMM showed that a naive regime model could collapse into one dominant out-of-sample state, while the weighted feature-block approach produced more interpretable regimes. The final regimes had clear differences in volatility, momentum, drawdowns, and CFTC/COT positioning behavior. These labels were then used to test whether the classification models performed better when given regime information directly or when trained separately within regimes. This connected the regime modeling step to the broader goal of the project: determining whether alternative data and market structure could improve WTI direction forecasting and trading performance.

6 Regime-Aware and Extended Experiments

After establishing the original four-model baseline, I extended the forecasting pipeline to test whether additional market information could improve WTI direction prediction. The original baseline models used price history and macroeconomic variables, but the baseline results showed that the daily directional signal was weak. I therefore introduced two main extensions: CFTC/COT trader-positioning data and HMM-based market regimes. The goal of this stage was not simply to add more variables, but to test whether alternative data and market-state information could provide a better structure for the forecasting problem.

For each of the four forecasting models, I evaluated four experimental versions: a pre-COT baseline, a COT-augmented version, a regime-probability version, and a regime-specific version. This design allowed me to separate the effect of adding trader-positioning data from the effect of adding regime information. It also allowed me to test whether regimes were more useful as an additional input feature or as a way to split the forecasting problem into separate market states.

Table 11: Extended experimental design.

Experiment Version	Feature / Modeling Setup	Purpose
Pre-COT baseline	Final sample period using market and macroeconomic features, but excluding CFTC/COT variables and HMM regime information.	Provides a comparison point for evaluating whether CFTC/COT positioning data adds value over the same date range.
COT-augmented model	Pre-COT feature set plus engineered CFTC/COT positioning variables, including net positioning, positioning percentages, flows, long-short ratios, open interest, and open-interest changes.	Tests whether trader-positioning information improves WTI direction prediction beyond market and macroeconomic predictors.
Regime-probability model	COT-augmented feature set plus the HMM regime probability columns as additional predictors. These probabilities represented the model’s estimated likelihood that each day belonged to each latent regime.	Tests whether giving the classifier probabilistic market-state information improves prediction while still using one unified model.
Regime-specific model	Separate models estimated within HMM regimes, allowing the predictor-target relationship to differ across market states.	Tests whether WTI direction prediction improves when different regimes are modeled separately rather than averaged into one global classifier.

The pre-COT version was important because the addition of CFTC/COT data changed the usable sample period. The original semester baseline used a longer historical macroeconomic dataset, while the COT-enhanced dataset began later because the merged CFTC/COT features were only available over the final combined sample. To make the extended experiments more comparable, I recreated a pre-COT version using the same final sample window as the COT-based models. This ensured that differences between the pre-COT and COT-augmented versions were driven by the feature set rather than by different date ranges. In the extended experiments, the training period covered 2015 through 2021, the validation period covered 2022 through 2023, and the test period covered 2024 through 2025.

In the extended experiments, I also modified how recent price history was represented. The original baseline framework used a 30-day lookback structure, but the final regime-aware models did not include the full raw sequence of 30 daily price lags. Instead, recent WTI price behavior was summarized through engineered market features, including daily returns, rolling volatility, momentum, drawdown, and trend-related variables. This design choice was consistent with the regime-aware extension of the project. Rather than allowing the classifiers to depend primarily on a high-dimensional raw lag structure, I represented recent WTI history through price-summary variables that were more directly connected to market states such as trending, volatile, or stressed environments. This made the final feature set more compact and interpretable while still preserving information about recent price dynamics. A limitation of this choice is that the final models may miss finer autoregressive patterns that a full lag-window specification could capture.

The COT-augmented version added engineered trader-positioning variables to the existing market and macroeconomic feature set. These variables included net positioning, net positioning as a percentage of open interest, changes in positioning flows, long-short ratios, total open interest, and changes in open interest. The purpose was to test whether trader behavior contained information that was not already captured by price history or macroeconomic indicators. For example, high non-commercial long positioning may reflect speculative bullishness, while sharp changes in positioning may reflect shifts in market expectations. Because

the CFTC/COT data is weekly while WTI prices are daily, the COT variables were aligned to the daily dataset and shifted to account for reporting timing before being merged into the forecasting pipeline.

The regime-probability version added the HMM regime probabilities directly as inputs to the classifier. Rather than using only the single most likely regime label, this approach preserved the HMM’s uncertainty about the market state on each day. For example, a day could be mostly associated with regime 0 while still having some probability of belonging to regime 1. This allowed the classifier to condition its prediction on a softer representation of the current market environment. The motivation was that the same feature value may have different implications depending on the probability of being in a stressed, stable, or event-driven regime. By including regime probabilities as features, the model could potentially learn how directional relationships changed with the estimated market state while still being trained as one unified classifier.

The regime-specific version used the HMM regimes in a more structural way. Instead of adding the regime probabilities as additional predictors in one unified classifier, I used the most likely HMM regime assignment to train separate models for different market states. This approach allowed the relationship between predictors and WTI direction to vary across regimes. The economic intuition was that oil markets may behave differently during stressed, volatile periods than during stable, positive-momentum periods. A single model may average across these relationships, while regime-specific models may capture them more directly by fitting separate decision rules within each market environment.

However, the regime-specific approach also introduced an important limitation. Splitting the data by the most likely regime assignment reduced the number of observations available to train each model. This created a trade-off between specialization and sample size. If the regimes truly represented different market environments, separate models could improve performance. But if a regime had too few observations, or if a regime did not appear in the test period, the regime-specific model could become unstable or less useful out of sample. This was especially relevant because regime 2 was concentrated in earlier years and did not appear in the final test period.

Model tuning was performed using the training and validation periods, while the final test period was held out for out-of-sample evaluation. For the final test evaluation, the classifiers were retrained on the combined training and validation data using the selected specifications, and predictions were generated on the held-out test set. This procedure preserved the chronological structure of the data and avoided tuning directly on the test period.

Overall, the extended experiment design allowed me to separate the effect of alternative data from the effect of regime information. The pre-COT and COT-augmented versions tested whether CFTC/COT positioning data improved the feature set, while the regime-probability and regime-specific versions tested whether HMM market-state information added useful structure.

7 Forecasting Results and Model Comparison

After defining the four extended experiment versions, I evaluated how each model performed on the held-out test period. This section focuses on statistical forecasting performance, while the following sections evaluate whether the predicted probabilities translated into economic value through a trading strategy. This distinction is important because a model can have only modest classification accuracy but still be useful for trading if its high-confidence predictions occur during economically meaningful price moves.

The final test results are reported in Table 12. Each model was evaluated across four versions: the pre-COT baseline, the COT-augmented model, the regime-probability model, and the regime-specific model. Accuracy, precision, recall, F1-score, and ROC-AUC were used to compare the models. As in the baseline section, the positive class is an up day.

Table 12: Final test classification results across models and experiment versions.

Model	Version	Accuracy	Precision	Recall	F1	ROC-AUC
Gradient Boosting	Pre-COT	0.502	0.501	0.722	0.592	0.519
Gradient Boosting	COT-Augmented	0.500	0.500	1.000	0.667	0.510
Gradient Boosting	Regime-Probability	0.521	0.511	0.983	0.672	0.520
Gradient Boosting	Regime-Specific	0.539	0.531	0.692	0.601	0.520
FAVAR	Pre-COT	0.500	0.500	0.759	0.603	0.513
FAVAR	COT-Augmented	0.500	0.500	0.738	0.596	0.501
FAVAR	Regime-Probability	0.500	0.500	0.755	0.602	0.501
FAVAR	Regime-Specific	0.520	0.514	0.755	0.612	0.536
GRU	Pre-COT	0.517	0.629	0.094	0.164	0.531
GRU	COT-Augmented	0.506	0.509	0.485	0.497	0.505
GRU	Regime-Probability	0.496	0.497	0.356	0.415	0.509
GRU	Regime-Specific	0.521	0.528	0.414	0.464	0.513
ELM	Pre-COT	0.502	0.501	0.764	0.605	0.501
ELM	COT-Augmented	0.494	0.497	0.983	0.660	0.519
ELM	Regime-Probability	0.521	0.512	0.928	0.660	0.512
ELM	Regime-Specific	0.503	0.502	0.975	0.663	0.524

Table 13: Selected final test confusion matrices illustrating classification behavior. Entries are reported as TN, FP, FN, TP, where the positive class is an up day.

Model	Version	TN	FP	FN	TP
Gradient Boosting	COT-Augmented	0	237	0	237
Gradient Boosting	Regime-Probability	14	223	4	233
Gradient Boosting	Regime-Specific	91	145	73	164
FAVAR	Regime-Specific	67	169	58	179
GRU	Pre-COT	218	13	211	22
GRU	Regime-Specific	142	84	133	94
ELM	COT-Augmented	1	236	4	233

Overall, the final test classification results remained modest across all models and experiment versions. Most accuracies were close to 50 percent, and ROC-AUC values were generally only slightly above random classification. This reinforces the difficulty of the forecasting task. Daily WTI direction is noisy, and many price movements are driven by short-term market news, inventory surprises, geopolitical developments, or changes in investor sentiment that may not be fully captured by macroeconomic variables, CFTC/COT positioning data, or HMM regimes.

The extended experiments are best interpreted as a sequence of incremental additions. The pre-COT version provides the starting point over the final sample period. The COT-augmented version tests whether trader-positioning data adds information. The regime-probability version tests whether soft HMM market-state information improves the unified classifier. The regime-specific version tests whether using regimes structurally, by fitting separate models within market states, adds value beyond simply adding regime information as predictors.

The effect of adding CFTC/COT data was mixed. COT data did not consistently improve hard classification accuracy. Gradient Boosting moved from 0.502 accuracy in the pre-COT version to 0.500 in the COT-augmented version, GRU declined from 0.517 to 0.506, and ELM declined from 0.502 to 0.494. However,

the COT features did change model behavior. For example, the Gradient Boosting COT-augmented model predicted every test observation as an up day, producing recall of 1.000 but zero true negatives. The ELM COT-augmented model showed a similar pattern, with 233 true positives but only one true negative. This suggests that COT information may have pushed some models toward an up-class bias rather than producing a balanced directional signal.

It is important to distinguish weak classification performance from class collapse. Metrics near 0.50 do not necessarily mean that a model collapsed into one class. For example, the FAVAR COT-augmented model had accuracy and precision near 0.50, but its confusion matrix still showed predictions in both classes, with 62 true negatives and 175 true positives. In contrast, the Gradient Boosting COT-augmented model predicted every observation as an up day, producing zero true negatives and zero false negatives. Therefore, the confusion matrices were necessary to distinguish models that were merely weak from models that had effectively collapsed into one prediction class.

The regime-probability versions also produced mixed but informative changes. Adding HMM regime probabilities improved Gradient Boosting accuracy from 0.500 to 0.521 and ELM accuracy from 0.494 to 0.521 relative to their COT-augmented versions. However, both models still showed strong up-class bias, with recall values of 0.983 and 0.928, respectively. FAVAR remained at 0.500 accuracy, and GRU fell below 50 percent accuracy in the regime-probability version. This suggests that soft regime probabilities helped some models, but adding regime information as another feature was not enough to consistently solve the classification problem.

The clearest incremental improvement came from the regime-specific versions. Gradient Boosting improved from 0.500 in the COT-augmented version and 0.521 in the regime-probability version to 0.539 in the regime-specific version. Its confusion matrix also became more balanced: instead of predicting every observation as up, it correctly classified 91 down days and 164 up days. FAVAR also performed best in the regime-specific version, reaching accuracy of 0.520, F1-score of 0.612, and the highest ROC-AUC in the table at 0.536. GRU also reached its best final accuracy in the regime-specific version at 0.521.

The GRU results show why the direction of class bias matters. Unlike Gradient Boosting and ELM, which often predicted too many up days, the GRU pre-COT model was too conservative about predicting up. It produced 218 true negatives but only 22 true positives, resulting in recall of only 0.094. The regime-specific GRU increased true positives from 22 to 94 and produced a more balanced confusion matrix. This suggests that regime-specific modeling helped the GRU recover some ability to identify up days, even though its overall classification performance remained poor.

The comparison also shows why ROC-AUC is important for this project. Accuracy depends on a fixed classification threshold, while ROC-AUC evaluates how well predicted probabilities rank up days above down days. In several cases, accuracy and ROC-AUC told different stories. For example, ELM's COT-augmented version had lower accuracy than its pre-COT version, but a higher ROC-AUC. This means the model's hard classifications were not better at the 0.50 threshold, but its probabilities contained somewhat better ranking information. Since the later trading strategy uses predicted probabilities rather than only hard class labels, ROC-AUC is especially relevant for evaluating whether a model may still contain useful signal.

Across the full set of results, the main takeaway is that CFTC/COT data and HMM regimes added structure, but they did not transform WTI direction prediction into a high-accuracy classification task. The incremental value was strongest when regime information was used structurally through regime-specific modeling rather than only as additional probability features. This pattern was most visible for Gradient Boosting, FAVAR, and GRU. However, the improvements were still modest, and several models showed class-bias issues. This motivated the next step of the project: evaluating whether weak probability signals could still produce economic value when translated into a risk-controlled trading strategy.

8 Trading Strategy Design

8.1 Motivation for Trading Evaluation

The forecasting models developed in this project were initially evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. While these metrics are useful for measuring statistical forecasting performance, they do not necessarily determine whether a model produces economically useful trading signals. In financial markets, the timing of correct and incorrect predictions, the confidence of the model, market volatility, and drawdowns all affect realized investment performance.

For this reason, the final stage of the project evaluated whether the forecast probabilities generated by the machine learning models could be translated into profitable trading decisions. Rather than treating every prediction equally, the trading framework used model confidence to determine both trading direction and position size. This was important because a prediction probability of 0.51 should not be treated the same as a prediction probability of 0.90. The trading framework was therefore designed to test whether stronger forecast confidence corresponded to more useful economic signals.

The trading strategies were not intended to represent a deployable live trading system. Instead, they were designed as a framework for evaluating whether the forecasting models contained economically meaningful directional information beyond standard classification metrics.

8.2 Probability-Based Trading Signals

The trading strategies were based on the predicted probabilities generated by each classification model. Each model produced a probability p_t representing the estimated probability that WTI crude oil prices would increase on the following trading day.

Instead of converting these probabilities directly into binary long or short positions, the framework used probability-based position sizing. The intuition behind this approach is that stronger prediction confidence should correspond to larger position sizes, while weaker prediction confidence should produce smaller or no positions.

The strategy first converts the predicted probability p_t into a raw directional signal:

$$s_t = \begin{cases} m(p_t - \theta_L) & \text{if } p_t \geq \theta_L \\ m(p_t - \theta_S) & \text{if } p_t \leq \theta_S \\ 0 & \text{otherwise} \end{cases}$$

where:

- m is the signal multiplier,
- θ_L is the long threshold,
- θ_S is the short threshold.

Probabilities above the long threshold generate positive signals, while probabilities below the short threshold generate negative signals. Predictions close to 0.5 generate little or no exposure because the models do not express strong directional confidence in those cases.

The raw signal is then scaled asymmetrically:

$$\tilde{s}_t = \begin{cases} \alpha s_t & \text{if } s_t \geq 0 \\ \beta s_t & \text{if } s_t < 0 \end{cases}$$

where:

- α controls the aggressiveness of long exposure,

- β controls the aggressiveness of short exposure.

This asymmetric scaling was useful because several models exhibited an upward directional bias, meaning they predicted upward price movements more frequently than downward movements.

Next, the signal is transformed using a nonlinear confidence parameter γ :

$$\hat{s}_t = \text{sign}(\tilde{s}_t) |\tilde{s}_t|^\gamma$$

The parameter γ controls how aggressively the strategy responds to different levels of forecast confidence. When $\gamma < 1$, smaller signals are amplified, making the strategy more responsive to modest probability deviations from the thresholds. When $\gamma > 1$, smaller signals are compressed and the strategy becomes more selective.

8.3 Volatility Targeting and Position Scaling

In addition to probability-based position sizing, the strategy also used volatility targeting to manage risk exposure. Crude oil markets are highly volatile and frequently experience periods of large price swings due to geopolitical events, macroeconomic shocks, supply disruptions, and changes in market sentiment. As a result, maintaining constant exposure across all market environments can produce unstable risk profiles.

To address this issue, the final trading position was scaled inversely with recent realized volatility:

$$w_t = \text{clip} \left(\hat{s}_t \cdot \frac{\sigma_{\text{daily}}^*}{\max(\sigma_{t,20}, \sigma_{\min})}, -w_{\max}, w_{\max} \right)$$

where:

- w_t is the final trading position,
- σ_{daily}^* is the target daily volatility,
- $\sigma_{t,20}$ is the rolling 20-day realized volatility estimate,
- $\sigma_{\min}$ is the volatility floor,
- $w_{\max}$ is the maximum allowable absolute position size.

The target daily volatility was computed from the annualized target volatility using:

$$\sigma_{\text{daily}}^* = \frac{\sigma_{\text{annual}}^*}{\sqrt{252}}$$

This framework reduces exposure during highly volatile periods while allowing larger effective positions during more stable market conditions. The volatility floor prevents unrealistically large leverage during periods of extremely low realized volatility.

8.4 Strategy Returns, Rebalancing, and Transaction Costs

The trading strategies were rebalanced daily using updated model predictions and volatility estimates. Daily rebalancing allowed the strategies to continuously incorporate new market information and changing forecast probabilities.

At each trading day t , the strategy first generated a trading position w_t based on the model probability forecast and volatility-scaling framework described earlier. The value of w_t represents the portfolio exposure to WTI crude oil returns during the following trading period. Positive values correspond to long exposure, negative values correspond to short exposure, and larger absolute values correspond to larger position sizes.

Strategy returns were calculated using the previous day’s position:

$$R_t^{\text{gross}} = w_{t-1} r_t$$

where:

- R_t^{gross} is the gross strategy return on day t ,
- w_{t-1} is the trading position held entering day t ,
- r_t is the realized WTI return during day t .

Using the lagged position ensures that the strategy only earns returns on positions that were determined before observing the current day’s return. This is important because using w_t directly would introduce look-ahead bias by allowing the strategy to react to information from the same period in which returns are measured.

For example, if the strategy entered day t with a 50% long position ($w_{t-1} = 0.5$) and WTI increased by 2% during that day ($r_t = 0.02$), then the gross strategy return would equal:

$$R_t^{\text{gross}} = 0.5 \times 0.02 = 0.01$$

or 1%.

Transaction costs were incorporated using a turnover penalty:

$$R_t^{\text{net}} = R_t^{\text{gross}} - c \cdot |w_t - w_{t-1}|$$

where:

- R_t^{net} is the net strategy return after transaction costs,
- c is the transaction cost parameter,
- $|w_t - w_{t-1}|$ measures portfolio turnover.

Turnover measures how much the portfolio position changes from one day to the next. Strategies with extremely high turnover may appear profitable before trading frictions are considered but become less realistic once transaction costs are included. For this reason, turnover was monitored during strategy tuning to discourage excessive trading activity and improve robustness.

8.5 Strategy Parameters and Hyperparameter Tuning

Several hyperparameters controlled the behavior of the trading strategies. These parameters were designed to balance responsiveness, risk control, and robustness.

The trading strategies were tuned using validation-period performance rather than the final test set. This was important to avoid overfitting the trading rules to the out-of-sample evaluation period.

The strategy hyperparameters were selected using grid-search experiments across multiple parameter combinations, including probability thresholds, position-scaling parameters, volatility-targeting strength, and target volatility levels. The tuning process evaluated trade-offs between total return, Sharpe ratio, drawdown, volatility, and turnover. The hyperparameters were tuned specifically on the validation set’s predictions on the 16 models from Section 7.

Table 14: Trading strategy parameters and interpretation.

Parameter	Interpretation
Signal multiplier (m)	Controls the magnitude of probability-based trading signals.
Long threshold (θ_L)	Minimum probability required before entering a long position.
Short threshold (θ_S)	Maximum probability required before entering a short position.
α	Controls the aggressiveness of long positions.
β	Controls the aggressiveness of short positions.
γ	Controls nonlinear confidence scaling of trading signals.
Target volatility	Desired annualized volatility level for the strategy.
Volatility floor	Prevents unrealistically large leverage during low-volatility periods.
Maximum position	Caps total exposure to limit excessive leverage.
Transaction cost	Penalizes turnover and approximates trading frictions.

Only after the strategy parameters were finalized using the validation data were the final strategies evaluated on the held-out test period. This separation between validation tuning and final testing was necessary to preserve the integrity of the out-of-sample evaluation.

The parameter sensitivity experiments generally produced smooth and structured changes in validation Sharpe ratios rather than isolated performance spikes. When plotting Sharpe against the tuned strategy parameters, the strongest performance appeared around a distinct and relatively smooth maximum. This suggested that the selected parameter values were part of a broader high-performing region, rather than a single unstable point. This increased confidence that the selected trading rule was not purely the result of overfitting to noise in the validation period.

Strategies with extremely high turnover were generally less stable after accounting for transaction costs. As a result, turnover was monitored during tuning to favor strategies with more realistic trading behavior and lower sensitivity to short-term prediction noise.

8.6 Validation Tuning Results

The final trading parameters were selected using validation-period predictions rather than test-period performance. This was important because the trading rule itself contains several parameters that could be overfit if chosen directly on the final test period. To reduce this risk, I tuned the strategy using the validation predictions from the different model and experiment-version combinations, then applied the selected parameters unchanged to the final test predictions.

Table 15: Validation-period performance for selected trading parameters.

Metric	Value
Average Sharpe	0.820
Median Sharpe	0.528
Average Total Return	0.239
Average Max Drawdown	-0.228
Worst Max Drawdown	-0.464
Average Absolute Position	0.457
Average Turnover	0.104

The validation results show that the selected trading parameters produced positive average performance across the validation prediction files, with an average Sharpe ratio of 0.820 and an average total return of 23.9%. However, the median Sharpe was lower at 0.528, and the worst maximum drawdown was -46.4%. This indicates that the strategy was not uniformly successful across all models and market conditions. This is important because it shows that the tuning process identified a parameter set with positive average validation performance, but it did not eliminate downside risk or guarantee strong performance for every model.

The validation tuning results also helped justify the use of risk controls in the strategy design. Average absolute position was 0.457, meaning the strategy did not simply take maximum exposure every day. Average turnover was 0.104, suggesting that the strategy adjusted positions regularly but did not rely on extreme daily position changes. These results supported the use of the selected probability thresholds, nonlinear scaling, volatility targeting, position clipping, and transaction cost assumptions before moving to the held-out test period.

Final parameters are shown in Table 16.

Table 16: Final Trading Parameters

Parameter	Value
Signal multiplier (m)	3.5
Long threshold (θ_L)	0.50
Short threshold (θ_S)	0.50
Long exposure scale (α)	3.0
Short exposure scale (β)	1.55
Confidence exponent (γ)	0.7
Target annual volatility	0.28
Volatility floor	0.005
Maximum absolute position	1.0
Transaction cost	0.0005

9 Trading Results and Evaluation

After evaluating the models statistically, I evaluated whether the forecast probabilities translated into economic value through the probability-based trading strategy. This section focuses on trading performance rather than classification performance. The motivation for this distinction is that a model with modest accuracy can still be economically useful if its probabilities are informative on larger return days, if its confidence is better aligned with market movement, or if the strategy reduces exposure during unfavorable market conditions. Conversely, a model with slightly stronger classification accuracy may still produce poor

trading results if its errors occur during large price moves or if its signals lead to excessive turnover and drawdowns.

9.1 Trading Evaluation Metrics

The trading strategies were evaluated using both return and risk metrics. Total return measures the cumulative return earned over the full test period. CAGR reports the compounded annual growth rate, which makes performance easier to compare across periods. Annualized volatility measures the variability of daily strategy returns scaled to a yearly frequency. The Sharpe ratio measures return per unit of volatility, so a higher Sharpe ratio indicates better risk-adjusted performance. Maximum drawdown measures the largest peak-to-trough decline in cumulative returns and is especially important for evaluating downside risk. I also report average absolute position and average turnover. Average absolute position measures how much exposure the strategy took on average, while turnover measures how much the position changed from day to day and therefore how sensitive the strategy may be to transaction costs.

9.2 Average Performance by Experiment Version

Before comparing individual models, I first summarized average trading performance across the four model families for each experiment version. This provides a high-level view of whether each stage of the extended pipeline added economic value. The pre-COT version provides the starting point over the final sample period. The COT-augmented version tests whether trader-positioning data improved trading performance. The regime-probability version tests whether soft HMM regime information helped when added as predictors. The regime-specific version tests whether using regimes structurally, by fitting separate models within market states, improved trading performance.

Table 17: Average final test trading performance across model families by experiment version.

Version	Total Return	CAGR	Ann. Vol	Sharpe	Max DD	Avg. Abs Pos	Avg. Turnover
Pre-COT	0.004	0.000	0.180	0.032	-0.204	0.525	0.151
COT-Augmented	-0.092	-0.050	0.207	-0.245	-0.302	0.671	0.261
Regime-Probability	-0.076	-0.042	0.200	-0.173	-0.242	0.621	0.243
Regime-Specific	0.143	0.072	0.199	0.416	-0.178	0.630	0.236

The average results show that the regime-specific version produced the strongest economic performance across model families. On average, regime-specific strategies generated a total return of 14.3%, a CAGR of 7.2%, and a Sharpe ratio of 0.416. They also had the smallest average maximum drawdown among the extended versions at -17.8%. This suggests that the most useful form of regime information was not simply adding regime probabilities as extra features, but using the regimes to structure the modeling problem.

The COT-augmented version performed worse on average than the pre-COT version. Average total return declined from 0.4% in the pre-COT version to -9.2% in the COT-augmented version, and average Sharpe declined from 0.032 to -0.245. This does not necessarily mean that COT data contained no useful information. Rather, it suggests that adding positioning variables directly to all models did not consistently improve the trading signal and, in some cases, increased exposure to poor signals. The regime-probability version improved on the COT-augmented version, with a less negative average return and Sharpe ratio, but it still remained negative on average. The clearest improvement came when regimes were used structurally through regime-specific models.

9.3 Model-Specific Trading Results

Table 18 reports selected model-level trading results from the final test period. I include the buy-and-hold benchmark, the strongest regime-specific strategies, and selected comparison models that help explain the

relationship between classification performance and trading value. The full table of results can be found in the appendix. The buy-and-hold benchmark represents a passive long-only position in WTI over the test period, while the model-based strategies use the probability-based trading rule described in the previous section.

Table 18: Selected final test trading results.

Model / Strategy	Version	Total Return	Sharpe	Max Drawdown
Buy-and-Hold	Benchmark	-0.207	-0.387	-0.381
Gradient Boosting	Pre-COT	0.016	0.061	-0.144
Gradient Boosting	Regime-Specific	0.071	0.387	-0.098
FAVAR	Pre-COT	-0.045	-0.110	-0.170
FAVAR	Regime-Specific	0.237	0.752	-0.133
GRU	Pre-COT	0.227	0.590	-0.208
GRU	Regime-Specific	0.420	0.840	-0.143
ELM	Regime-Probability	-0.081	-0.194	-0.328
ELM	Regime-Specific	-0.156	-0.318	-0.340

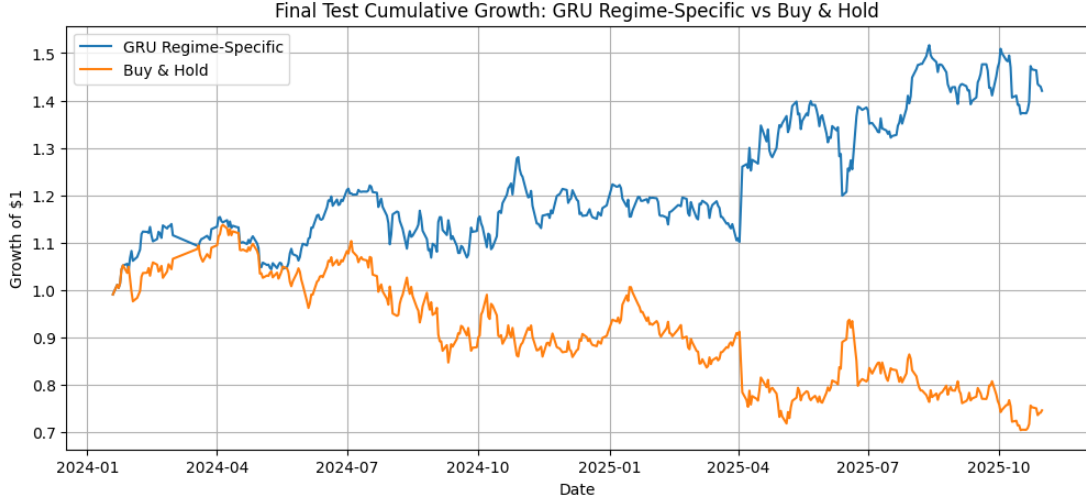
The buy-and-hold benchmark performed poorly during the final test period, with a total return of approximately -20.7%, a negative Sharpe ratio, and a maximum drawdown of approximately -38.1%. This benchmark is important because it shows that simply holding WTI exposure during the test period was not profitable and involved substantial downside risk. Therefore, a model-based strategy did not need to generate extremely high returns to be useful; reducing losses and drawdowns relative to buy-and-hold was also economically meaningful.

The model-specific results show that regime-specific modeling was especially useful for several model families. For Gradient Boosting, the regime-specific version improved total return from 1.6% in the pre-COT version to 7.1%, while reducing maximum drawdown from -14.4% to -9.8%. This was not the highest-return strategy, but it was one of the strongest risk-control results. The Gradient Boosting regime-specific strategy suggests that regime structure helped the model avoid some unfavorable exposure, even though its classification accuracy remained modest.

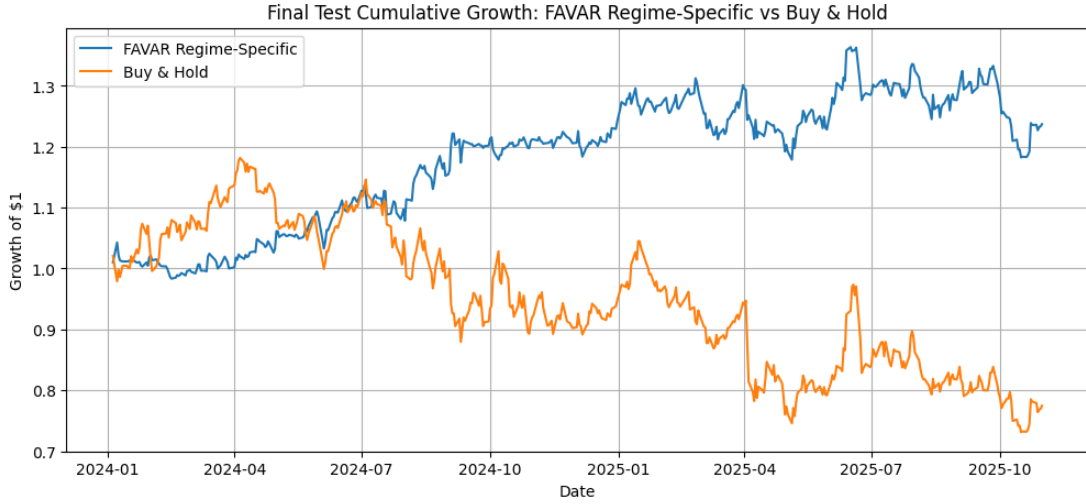
FAVAR also benefited substantially from regime-specific modeling. The FAVAR pre-COT strategy lost 4.5% with a negative Sharpe ratio, while the FAVAR regime-specific strategy gained 23.7% with a Sharpe ratio of 0.752. This is one of the most coherent results in the project because the FAVAR regime-specific model also had the strongest ROC-AUC in the final forecasting results. This suggests that the macro-factor model produced more useful probabilities when the relationship between predictors and WTI direction was allowed to vary across regimes.

The GRU results provide the strongest example of the difference between statistical accuracy and trading value. The GRU pre-COT classifier had very low recall and identified few up days, but its trading strategy still generated a 22.7% total return and a Sharpe ratio of 0.590. The GRU regime-specific strategy performed even better, producing the strongest overall trading result with a 42.0% total return and a Sharpe ratio of 0.840. Its maximum drawdown was -14.3%, much smaller than the buy-and-hold drawdown despite taking relatively high exposure. This suggests that the GRU probabilities contained economically useful information even though the hard classification metrics were only modest.

The ELM strategies provide a cautionary comparison. In the classification results, several ELM versions had high recall and F1-scores because they predicted up days frequently. However, this did not translate into strong trading performance. Both the ELM regime-probability and regime-specific strategies lost money, with negative Sharpe ratios and drawdowns larger than the stronger regime-specific strategies. This reinforces the point that high recall or similar classification accuracy does not necessarily imply economic usefulness.



(a) GRU Regime-Specific



(b) FAVAR Regime-Specific

Figure 3: Final test cumulative growth for the GRU and FAVAR regime-specific strategies compared with buy-and-hold. GRU produced the strongest total return and Sharpe ratio, while FAVAR provided the clearest link between improved probability ranking and trading performance.

The full set of model-version trading results is reported in Appendix Table 21.

9.4 Classification Accuracy vs. Trading Value

The trading results show that higher classification accuracy did not necessarily imply better trading performance. For example, Gradient Boosting regime-specific had the highest classification accuracy in Table 12, but it did not produce the highest trading return or Sharpe ratio. Meanwhile, GRU regime-specific had only modest classification accuracy, yet it produced the strongest trading performance. Similarly, ELM regime-probability had the same test accuracy as GRU regime-specific, but its trading result was negative.

This difference occurs because classification metrics treat each day equally, while trading performance depends on the size and timing of returns. Correctly predicting a small price move has less economic value than correctly positioning before a larger move. Similarly, incorrect predictions during high-volatility periods

can be much more damaging than incorrect predictions during calm periods. The trading strategy also used predicted probabilities, not just hard class labels. As a result, the probability distribution, confidence level, volatility environment, and position-sizing rule all affected realized performance.

The results therefore support one of the main lessons of the project: statistical forecasting performance and economic trading value are related, but they are not the same. Accuracy, precision, recall, F1-score, and ROC-AUC are useful for understanding model behavior, but they do not fully determine whether a model can generate useful trading signals. In this project, the best economic results came from models whose probabilities interacted well with the trading strategy, especially the regime-specific GRU and FAVAR models.

9.5 Key Takeaways

The main takeaway from the trading evaluation is that the model-based strategies were able to extract some economic value from weak statistical signals. The buy-and-hold benchmark lost money and experienced a large drawdown during the final test period, while several model-based strategies produced positive returns and smaller drawdowns. The strongest results came from the regime-specific GRU and FAVAR strategies, suggesting that using market regimes structurally was more useful economically than it appeared from classification accuracy alone.

The average performance across model families supports the same conclusion. The regime-specific version was the only experiment version with clearly positive average return and Sharpe ratio across all four model families. In contrast, directly adding COT features or regime probabilities did not consistently improve trading performance on average. This suggests that the regime framework added the most value when it changed the structure of the modeling problem rather than simply expanding the feature set.

The results also show the importance of risk controls. Volatility targeting, position clipping, and transaction costs helped make the strategy evaluation more realistic than a simple long/short rule based only on predicted class labels. The Gradient Boosting regime-specific strategy is a good example: its total return was lower than the GRU and FAVAR strategies, but it substantially reduced drawdown relative to buy-and-hold.

Overall, the trading results suggest that CFTC/COT data and HMM regimes did not create highly accurate daily classifiers, but they did contribute to a more useful forecasting-and-trading pipeline. The strongest evidence came from the regime-specific strategies, which showed that weak probability signals could still be economically meaningful when combined with position sizing and risk management.

10 Discussion

The results of this project show that forecasting daily WTI crude oil direction remains difficult even after adding macroeconomic predictors, CFTC/COT futures positioning data, HMM regime modeling, and multiple machine learning approaches. Across the original baseline models and the extended model versions, classification performance generally stayed close to random prediction. This does not mean that the models learned no information, but it does show that daily oil direction is a noisy target affected by short-term news, inventory surprises, geopolitical events, market sentiment, and supply-demand expectations that are difficult to capture with structured macroeconomic and positioning variables.

10.1 Main Findings

The original macroeconomic and price-history models established the difficulty of the forecasting task. FAVAR and Gradient Boosting were the strongest original baselines, ELM produced more balanced but weak predictions, and the original GRU collapsed into predicting one class. These results showed that model complexity alone was not enough to overcome the low signal-to-noise ratio of daily WTI direction prediction.

Adding CFTC/COT positioning data provided an economically meaningful alternative data source, but it did not automatically improve forecasting performance. The COT variables captured trader positioning, open

interest, and speculative imbalance, which are different from the information contained in macroeconomic variables. However, the COT-augmented models often showed mixed results and sometimes became more biased toward predicting up days. This suggests that COT data may contain useful market information, but that information is noisy and does not translate directly into better daily classification when added as a standard feature block.

The most consistent improvement came from using HMM regimes structurally. Adding regime probabilities as features helped some models, but the regime-specific versions produced the clearest gains in both classification balance and trading performance. This suggests that the value of regime modeling came less from adding another predictor and more from allowing predictor relationships to differ across market environments. The HMM regimes were not perfect economic labels, but they captured meaningful differences in volatility, momentum, drawdowns, and CFTC/COT positioning behavior.

10.2 Forecasting Accuracy versus Economic Value

One of the most important lessons from the project is that statistical forecasting performance and trading performance are related but not identical. Classification metrics evaluate whether the model predicts the correct direction on each day, while trading performance depends on the size and timing of returns, the model's confidence, the volatility environment, transaction costs, and position sizing.

This distinction was clear in the final results. Gradient Boosting regime-specific had the strongest classification accuracy, but it did not produce the highest trading return. GRU regime-specific had only modest classification accuracy, yet it produced the strongest trading performance. ELM also showed that high recall or frequent up-day predictions did not necessarily translate into economic value. These comparisons show why probability quality and signal timing matter more for trading than hard classification accuracy alone.

The strongest economic results came from the regime-specific GRU and FAVAR strategies. The GRU regime-specific strategy produced the highest total return and Sharpe ratio, while the FAVAR regime-specific strategy provided the clearest connection between improved probability ranking and trading performance. Gradient Boosting regime-specific was also useful from a risk-control perspective because it reduced drawdown relative to buy-and-hold. Overall, the trading results suggest that weak statistical signals can still have economic value when combined with probability-based position sizing, volatility targeting, and regime-aware model structure.

10.3 Limitations

There are several limitations to this project. First, the target variable is inherently noisy. Daily WTI direction is affected by many short-term events that are difficult to capture using monthly macroeconomic data, weekly positioning data, and engineered price features.

Second, the final COT-augmented dataset had a shorter sample period than the original macroeconomic dataset. Adding COT data introduced a potentially useful alternative signal, but it reduced the amount of historical data available for model training. This trade-off was especially important for more complex models and for regime-specific models, where splitting the data by regime further reduced the number of observations available within each state.

Third, the HMM regimes were data-driven and should not be interpreted as perfect economic states. Although the regimes had meaningful differences in volatility, momentum, drawdowns, and positioning behavior, they were estimated statistically. The fact that regime 2 disappeared in the final test period shows that some regimes may capture historically specific environments rather than stable recurring market states.

Fourth, the trading strategy was a simplified research framework rather than a production-ready trading system. It included transaction costs and volatility scaling, but it did not fully model bid-ask spreads, slippage, liquidity constraints, margin requirements, contract roll costs, or execution timing. These real-world frictions could reduce performance if the strategy were implemented live.

Finally, the trading strategy itself had several hyperparameters. Although the parameters were selected using validation data rather than the final test set, there is still a risk that the trading rule was partially fit to the validation period. The smooth validation Sharpe patterns helped reduce this concern, but future work should use a more formal walk-forward evaluation to test whether the selected strategy remains stable across multiple out-of-sample periods.

11 Conclusion and Future Work

This project examined whether daily WTI crude oil price direction could be forecast using macroeconomic indicators, CFTC/COT futures positioning data, HMM regime modeling, and four machine learning classification models: Gradient Boosting, ELM, GRU, and FAVAR. The results showed that daily WTI direction is difficult to predict. Across both the original baseline models and the extended experiments, classification accuracy generally remained close to 50 percent, and ROC-AUC values were only slightly above random prediction.

The CFTC/COT data added a useful alternative source of information because it captured futures-market positioning rather than only macroeconomic conditions or price history. However, adding COT features directly did not consistently improve classification performance. In some cases, it increased up-class bias rather than producing a more balanced directional signal. The HMM regime extension was more useful when it changed the structure of the modeling problem. The regime-specific models produced the clearest incremental gains, suggesting that predictor relationships may differ across market environments.

The trading evaluation showed that statistical forecasting performance and economic value are related but not identical. Several models had only modest classification accuracy but still produced useful trading signals when their probabilities were translated into volatility-scaled positions. The strongest trading results came from the regime-specific GRU and FAVAR strategies, which outperformed buy-and-hold during the final test period. This suggests that even weak probability signals can have economic value when combined with position sizing, risk management, and regime-aware model structure.

There are several directions for future work. First, I would develop a regime-aware trading rule that directly incorporates regime information into thresholds, position sizing, volatility targets, or risk limits. In this project, I used the same trading strategy across all model versions to keep the comparison fair, but this meant the strategy itself did not adapt to regimes. Second, I would estimate HMM regimes on the original no-COT dataset and rerun the regime-probability and regime-specific models without COT variables. This would separate the value of regime modeling from the value of COT alternative data.

Additional future work could compare the engineered price-summary features against a full 30-day raw price-lag specification, improve probability calibration before trading, and use a formal walk-forward evaluation across multiple out-of-sample periods. I would also make the trading simulation more realistic by incorporating bid-ask spreads, slippage, contract roll costs, margin requirements, and liquidity constraints. Overall, the project shows that alternative data and regime modeling did not make daily WTI direction easy to classify, but they did help create a more informative forecasting-and-trading pipeline.

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A Appendix

Table 19: Full final test confusion matrices across models and experiment versions. Entries are reported as TN, FP, FN, TP, where the positive class is an up day.

Model	Version	TN	FP	FN	TP
Gradient Boosting	Pre-COT	67	170	66	171
Gradient Boosting	COT-Augmented	0	237	0	237
Gradient Boosting	Regime-Probability	14	223	4	233
Gradient Boosting	Regime-Specific	91	145	73	164
FAVAR	Pre-COT	57	180	57	180
FAVAR	COT-Augmented	62	175	62	175
FAVAR	Regime-Probability	58	179	58	179
FAVAR	Regime-Specific	67	169	58	179
GRU	Pre-COT	218	13	211	22
GRU	COT-Augmented	122	109	120	113
GRU	Regime-Probability	147	84	150	83
GRU	Regime-Specific	142	84	133	94
ELM	Pre-COT	57	180	56	181
ELM	COT-Augmented	1	236	4	233
ELM	Regime-Probability	27	210	17	220
ELM	Regime-Specific	8	229	6	231

Table 20: Trading strategy hyperparameter search space.

Parameter	Values Tested
Signal multiplier	0.5, 0.75, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0
Long threshold	0.40, 0.45, 0.50, 0.55, 0.60
Short threshold	0.60, 0.55, 0.50, 0.45, 0.40
Alpha	0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.5
Beta	0.5, 0.75, 1.0, 1.10, 1.20, 1.25, 1.30, 1.35, 1.4, 1.45, 1.5, 1.55, 1.60, 1.65, 1.70, 1.75
Gamma	0.4, 0.45, 0.5, 0.55, 0.60, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 1.0, 1.3
Target annual volatility	0.10, 0.15, 0.20, 0.22, 0.24, 0.25, 0.26, 0.28, 0.30, 0.32, 0.34, 0.35
Maximum position	1.0
Transaction cost	0.0005

Across the staged tuning process, the cumulative search space covered 9,734,400 possible parameter combinations, although the full Cartesian grid was not run in a single pass.

Table 21: Full final test trading results across models and experiment versions.

Model	Version	Total Return	CAGR	Ann. Vol	Sharpe	Max DD	Avg. Pos	Avg. Abs Pos	Avg. Turnover
Gradient Boosting	Pre-COT	0.016	0.008	0.136	0.061	-0.144	0.273	0.379	0.123
Gradient Boosting	COT-Augmented	-0.082	-0.044	0.165	-0.269	-0.228	0.535	0.535	0.037
Gradient Boosting	Regime-Probability	-0.044	-0.024	0.181	-0.132	-0.191	0.528	0.534	0.057
Gradient Boosting	Regime-Specific	0.071	0.037	0.097	0.387	-0.098	0.202	0.289	0.073
FAVAR	Pre-COT	-0.045	-0.024	0.138	-0.110	-0.170	0.304	0.410	0.366
FAVAR	COT-Augmented	-0.087	-0.047	0.132	-0.303	-0.206	0.293	0.403	0.382
FAVAR	Regime-Probability	-0.023	-0.012	0.132	-0.027	-0.202	0.296	0.395	0.366
FAVAR	Regime-Specific	0.237	0.120	0.170	0.752	-0.133	0.378	0.577	0.455
GRU	Pre-COT	0.227	0.118	0.200	0.590	-0.208	-0.545	0.602	0.067
GRU	COT-Augmented	-0.071	-0.039	0.255	-0.154	-0.471	0.005	0.863	0.581
GRU	Regime-Probability	-0.156	-0.088	0.260	-0.338	-0.248	-0.218	0.842	0.501
GRU	Regime-Specific	0.420	0.216	0.257	0.840	-0.143	-0.138	0.785	0.359
ELM	Pre-COT	-0.181	-0.101	0.244	-0.414	-0.293	0.563	0.709	0.048
ELM	COT-Augmented	-0.127	-0.070	0.275	-0.254	-0.304	0.775	0.885	0.043
ELM	Regime-Probability	-0.081	-0.044	0.228	-0.194	-0.328	0.680	0.713	0.050
ELM	Regime-Specific	-0.156	-0.087	0.273	-0.318	-0.340	0.853	0.868	0.057

Table 22: Top Five Validation Strategy Parameter Combinations by Average Sharpe Ratio

Signal Mult.	Long Thresh.	Short Thresh.	Alpha	Beta	Gamma	Target Ann. Vol	Avg. Sharpe	Avg. Total Return	Worst Max DD	Avg. Abs. Pos.	Avg. Turnover
3.5	0.50	0.50	3.00	1.55	0.70	0.28	0.819	0.239	-0.464	0.457	0.103
3.0	0.50	0.50	2.50	1.30	0.70	0.35	0.819	0.237	-0.465	0.454	0.103
3.0	0.50	0.50	2.50	1.35	0.70	0.35	0.819	0.240	-0.464	0.457	0.104
3.5	0.50	0.50	3.00	1.60	0.70	0.28	0.819	0.242	-0.464	0.459	0.104
3.0	0.50	0.50	3.00	1.45	0.65	0.32	0.819	0.254	-0.463	0.475	0.108